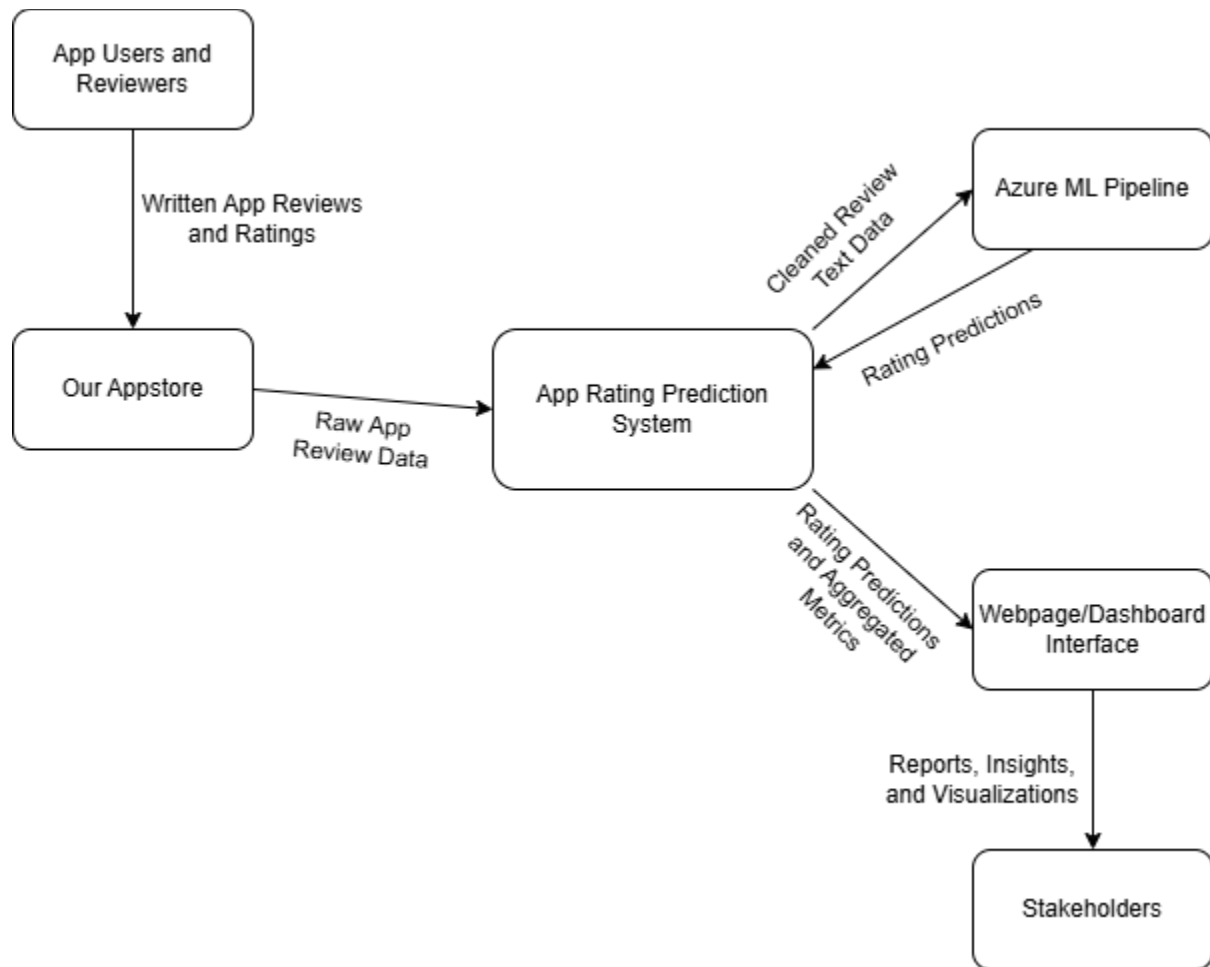
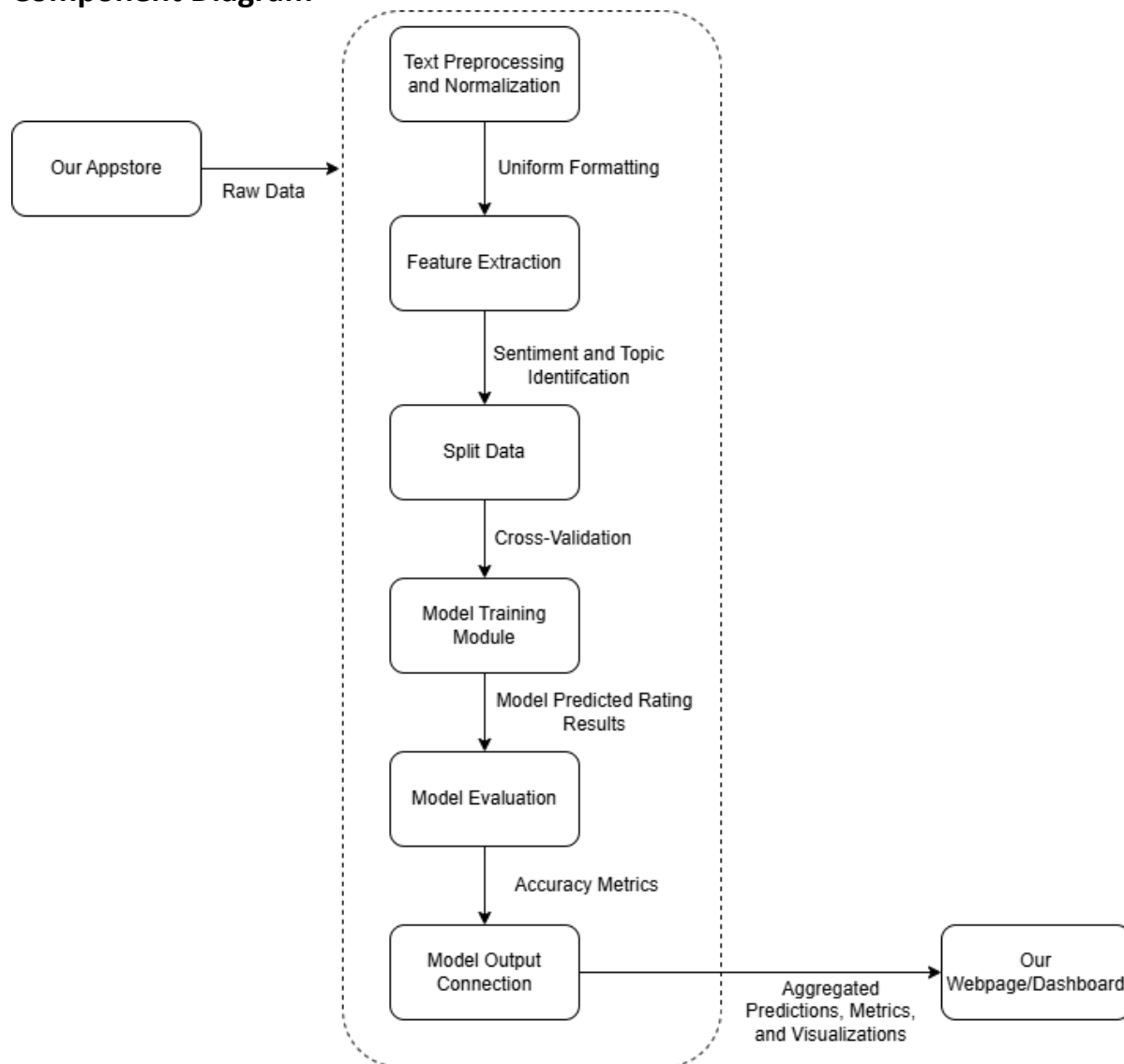


Context Diagram



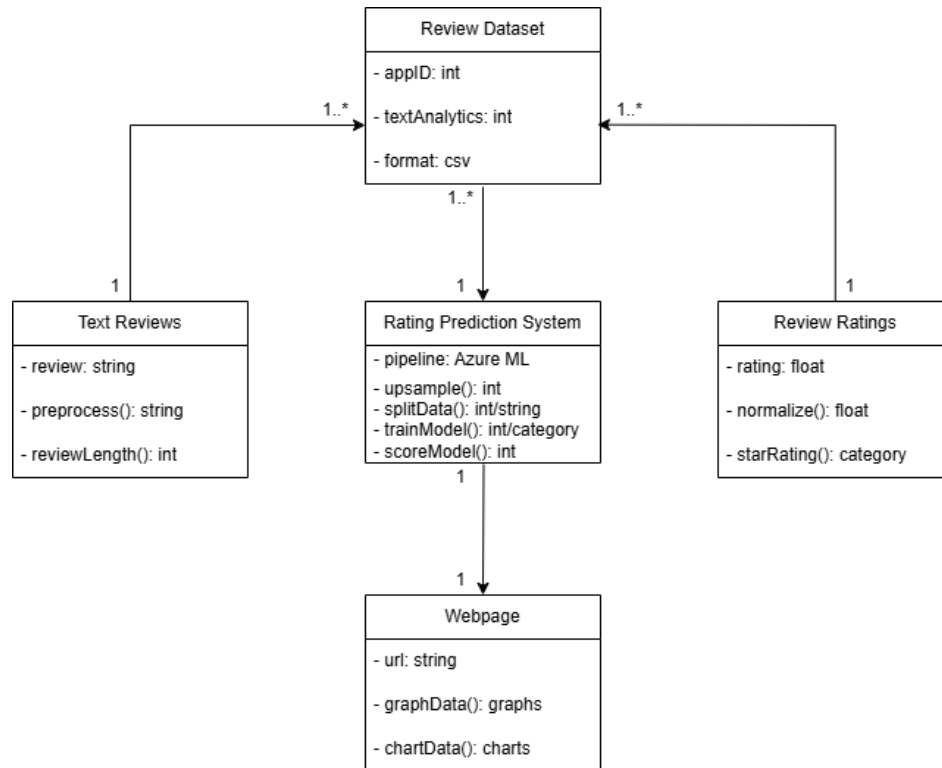
This context diagram shows how our App Rating Prediction System as a single entity interacts with all the external actors. Users will write app reviews and rate apps in our Appstore interface, and those free-text reviews will serve as the input for the prediction system. Those raw reviews will go into our App Rating Prediction System which works through an Azure ML pipeline which preprocesses and splits the data before training a model to predict an app's given rating based on the written review and finally the accuracy of the model is measured. Once all the data has gone through the prediction system, that is where it will get aggregated and sent to a connected dashboard which will be linked to our website which serves as an interface for our stakeholders and end users. This will provide stakeholders with rating predictions and other aggregated insights to support analysis and decision-making.

Component Diagram



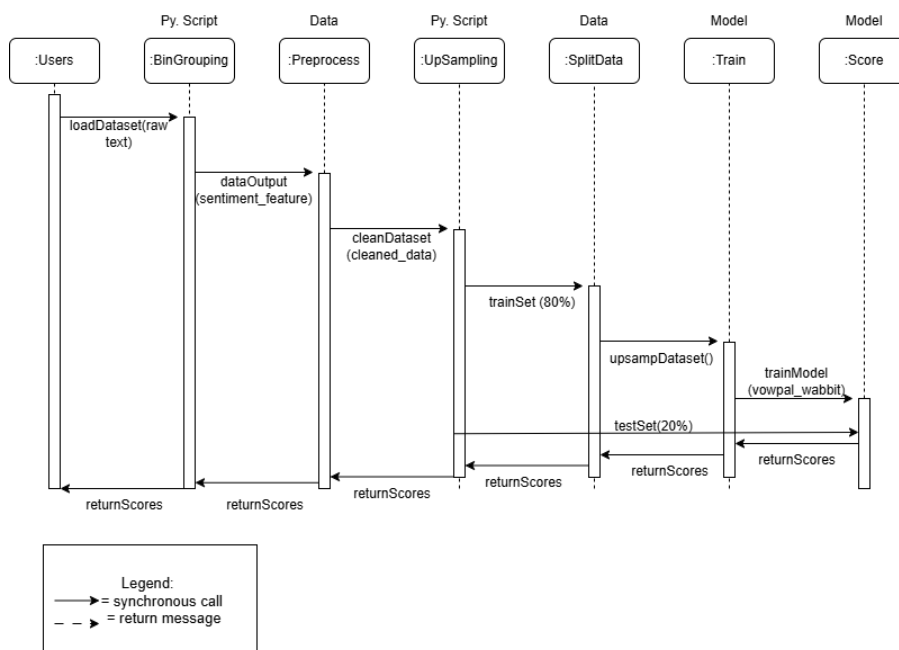
This component diagram shows the high-level flow of our app rating prediction system within an Azure ML pipeline. It starts with raw text review data is sent and input into our rating system. The data will first go through data cleaning and preprocessing, getting them in a standard format so that they can be used in the model. That processed data is then analyzed for its sentiment, tone, and topic to aid in the rating prediction. The processed data is then split using cross-validation and fed into the prediction models, which get trained and then tested to predict an app's rating from its reviews. The generated output predictions are stored and then once all the reviews have been predicted, accuracy metrics are calculated based on those predictions. Those predictions and accuracy metrics are then sent to a connected dashboard which is linked to a webpage to display them as aggregated insights in the form of data tables, charts, graphs, and other visualizations.

Class Diagram



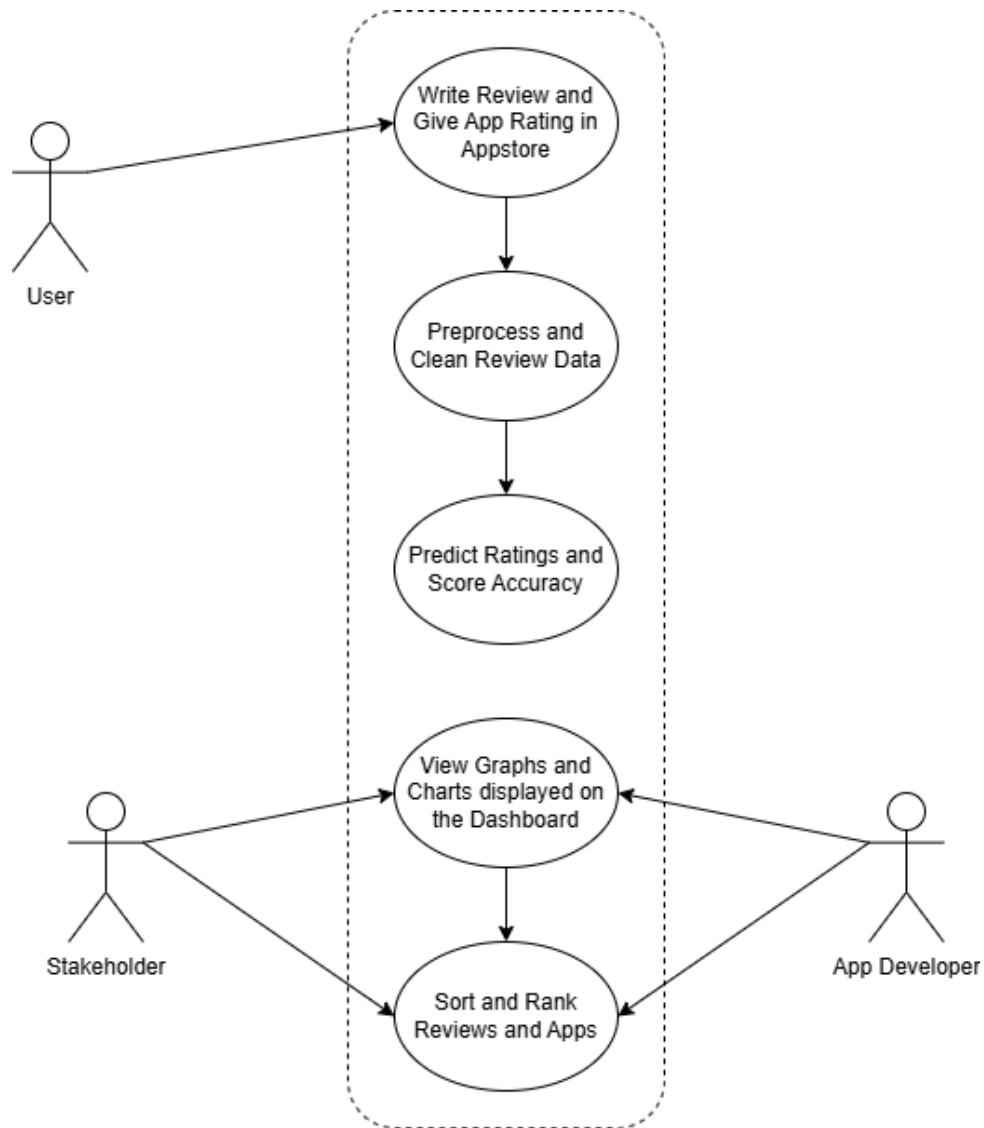
Our class diagram further displays the core entities of the system, both internal and external, including classes like the text reviews and ratings, along with their attributes and relationships, showing how the system interacts with review text data and how results are achieved and returned to the user.

Sequence Diagram



Our sequence diagram then traces this entire flow from start to finish, highlighting the order of interactions between components such as the UI and the ML pipeline at each step.

Use Case Diagram



The use case diagram captures the interactions between the main actors, like the stakeholders, app developers, and app reviewers, and the system's key functionalities, all the way from receiving a submitted app review, determining a predicted rating, and managing the prediction model. It provides a high-level view of how the system interacts with Users at each step.